

## EX. NO: 3 — HILL CIPHER

### Aim:

To implement the Hill Cipher substitution technique using C.

### Algorithm:

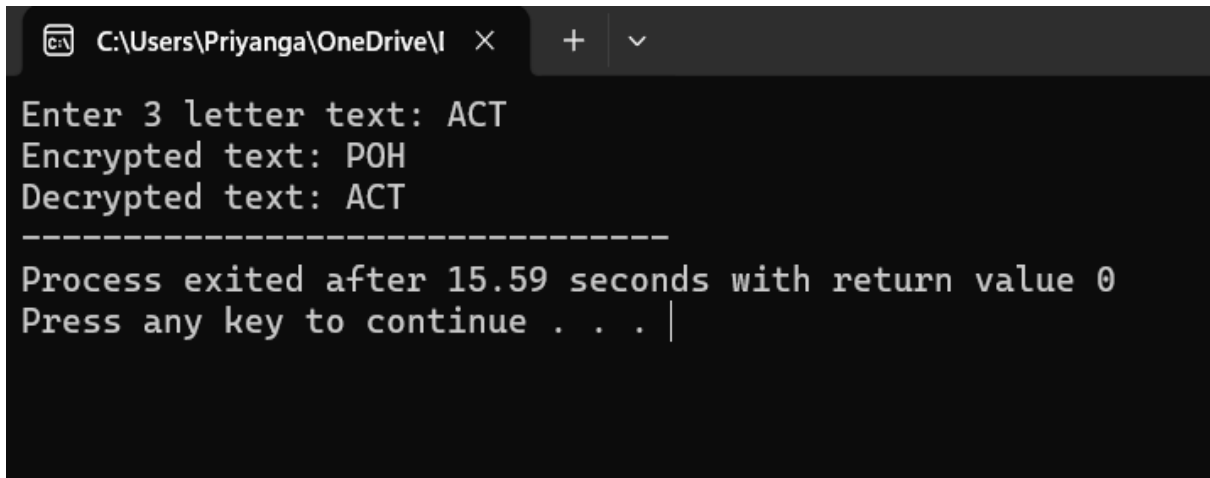
1. Read three-letter plaintext.
2. Convert letters into numbers.
3. Multiply plaintext with the  $3 \times 3$  key matrix.
4. Apply modulo 26.
5. Display encrypted text.
6. Decrypt using inverse matrix.

### Program:

```
#include <stdio.h>
#include <ctype.h>
int main()
{
    char text[4];
    int key[3][3] = {
        {6,24,1},
        {13,16,10},
        {20,17,15}
    };
    int inv[3][3] = {
        {8,5,10},
        {21,8,21},
        {21,12,8}
    };
    int p[3], c[3], d[3];
    int i, j, sum;
```

```
printf("Enter 3 letter text: ");
scanf("%3s", text);
for(i = 0; i < 3; i++)
    p[i] = toupper(text[i]) - 'A';
for(i = 0; i < 3; i++)
{
    c[i] = 0;
    for(j = 0; j < 3; j++)
        c[i] += key[i][j] * p[j];
    c[i] %= 26;
}
printf("Encrypted text: ");
for(i = 0; i < 3; i++)
    printf("%c", c[i] + 'A');
for(i = 0; i < 3; i++)
{
    d[i] = 0;
    for(j = 0; j < 3; j++)
        d[i] += inv[i][j] * c[j];
    d[i] %= 26;
}
printf("\nDecrypted text: ");
for(i = 0; i < 3; i++)
    printf("%c", d[i] + 'A');
return 0;
}
```

### Output:

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Priyanga\OneDrive\I' and standard window controls. The command prompt displays the following text: 'Enter 3 letter text: ACT', 'Encrypted text: POH', 'Decrypted text: ACT', followed by a line of dashes '-----'. Below the dashes, it says 'Process exited after 15.59 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

```
C:\Users\Priyanga\OneDrive\I > Enter 3 letter text: ACT
Encrypted text: POH
Decrypted text: ACT
-----
Process exited after 15.59 seconds with return value 0
Press any key to continue . . . |
```

### Result:

Thus, the Hill Cipher substitution technique was implemented successfully using C.